UNIVERSITAS INDONESIA

DO INDUSTRIES LEARN FROM UPSTREAM? THE ROLE OF VERTICAL SPILLOVERS IN SHAPING ENERGY INTENSITY ACROSS ASEAN-5 COUNTRIES

UNDERGRADUATE THESIS

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CHAPTER 1
INTRODUCTION

1.1 Background

Over the course of the mid-18th century, the invention of early mechanical technologies fundamentally transformed economies from artisanal production to mechanized manufacturing. What began as the first Industrial Revolution in Great Britain soon radiated across Western Europe, then to North America, before transforming economies across the globe (Landes, 1969). Although numerous countries in the Global South industrialized later in the twentieth century, the core features established during this period continue to define how production is organized today (Mokyr, 1990).

Among the major transformations of this revolution was a profound shift in how energy was generated and utilized, manifested in the adoption of coal-based power, steam technology, electrification, petroleum fuels, and internal-combustion engines. Once industries acquired a coal base, they were free to expand to a degree previously impossible. The transition to coal-based energy removed prior production limits, enabling significant mass production in industries such as salt, paper, glass, non-ferrous metals, and iron (Britannica, 2026). Between 1720 and 1850 alone, British iron production surged from around 25,000 tons to 1.4 million tons, marking a fifty-six-fold rise (Allen, 2009). This shift represents what Wrigley (2010) defines as the transition from an 'organic economy' to a 'mineral-based energy economy,' where the adoption of fossil fuels allowed humanity to bypass the biological limits of production.

However, such expansion was never without consequence. As Grossman and Krueger (1995) illustrate, this early stage of industrialization was dominated by a massive 'scale effect,' where the sheer volume of output growth far outpaced any nascent technological efficiency. Two centuries later, the world is paying its due with unprecedented environmental degradation. The 'safe operating space' for humanity is shrinking (Rockström et al., 2009), as mounting environmental repercussions impose growing physical and regulatory constraints on continued industrial expansion.

This environmental deadlock necessitates a fundamental shift in the logic of industrial development. Today, the global production system is being reconfigured by

the imperatives of climate transition; the pursuit of growth is no longer a matter of expanding volume, but of optimizing efficiency (Humphrey & Sturgeon, 2005; Marchi et al., 2013; Ward et al., 2016). Achieving such efficiency depends critically on technological transformation, especially the adoption and absorption of low-carbon technologies (Acemoglu et al., 2012; Jaffe et al., 2002).

Conceptually, low-carbon technology is defined as a replacement for "brown" (fossil-based) technologies with technologies that minimize carbon emissions along the production chain (Acemoglu et al., 2012). Popp et al. (2010) highlighted that this technology is not merely a new product, but often involves the improvement of existing machinery to make it more energy-efficient. The practical success of industrial upgrading is most clearly reflected in the trajectory of energy intensity, which serves as a broad economic measure of an industry's energy productivity. While technical efficiency focuses on the physical output per unit of energy, energy intensity provides a broader economic measure, namely through the amount of energy required to generate one unit of value added (Filippini & Hunt, 2015).

Empirical analyses reveal that the primary catalyst for these reductions is technological improvement and intra-sectoral efficiency gains, while broad structural economic shifts play a largely secondary role across most major economies (Voigt et al., 2014; Chang et al., 2019). Despite this overarching global decline, the dynamics of this energy transition remain highly dependent on national income levels; middle-income countries demonstrate rapid convergence and tend to form distinct convergence clubs, whereas the trajectories of high-income economies may actually diverge as they approach different technological frontiers (Chang et al., 2019).

Although the drivers of energy intensity varies, efficiency improvements are generally most affected by technological and structural shifts (Shakya et al., 2022; Eder & Provornaya, 2018; Chang et al., 2019). In this sense, energy intensity serves as a measurable manifestation of technological upgrading within a production system. A decline in intensity indicates that an industry is becoming more 'energy-productive,' successfully decoupling its economic growth from its total energy demand. These determinants interact in complex ways, manifesting distinct temporal and regional convergence patterns (Kazmaganbetova et al., 2016; Shakya et al., 2022; Wang et al., 2020), which are ultimately reflected in broader macro-level trends.

At the global level, energy intensity has persistently exhibited exponential downward trends across recent decades (Eder & Provornaya, 2018). Empirically, this pattern is most evident in advanced economies, where industrial energy intensity has generally declined over recent decades. For example, the energy intensity of the industrial sector in the European Union fell by around 27% between 2000 and 2020 (Tsemekidi Tzeiranaki et al., 2022). Similarly, most manufacturing subsectors in OECD countries also exhibit negative energy intensity growth rates over 1970–2005, most of which are driven primarily by efficiency improvements within sectors (Mulder & Groot, 2012). Past studies show that within-sector improvements are achieved through technological innovations, often proxied through patents (Rahko, 2025; Popp, 2001; Wurlod & Noailly, 2018; Ajayi and Reiner, 2020).

Despite this overarching global decline, Chang et al. (2019) noted that the dynamics of this energy transition remain highly dependent on national income levels. Developing and emerging economies are rapidly improving their energy efficiency, effectively catching up to global standards, but they aren't all following one single path. Instead, they group together into "clubs" based on similar industrial structures, policies, or regions. On the other hand, the wealthiest countries have already picked all the "low-hanging fruit" when it comes to saving energy. Since they are already highly efficient, they have to invent or adopt brand-new technologies, making them diverge.

This dynamic underscores the reality that the experience of developing economies is far less straightforward. The mechanisms that drive energy intensity reductions in advanced economies, including frontier technologies, strong regulatory frameworks, and shifts toward services, tend to function more weakly or differently in contexts with limited technological capabilities (Azaliah et al., 2024). In Southeast Asia, for instance, long-run trends in energy intensity reflect tension between structural change and limited efficiency gains. A decomposition analysis of six ASEAN economies over 1971–2016 reveals that all ASEAN countries shifted toward manufacturing and transport, sectors that are inherently more energy-intensive. In 2016, total final energy consumption across ASEAN-6 reached about 276 Mtoe, with manufacturing and transportation accounting for roughly 80% of the total (Setyawan, 2020). While declines in within-industry energy intensity have partially offset structural effects and reduced aggregate energy intensity over recent decades, efficiency gains remain modest by

international standards. OECD/IEA (2013) reports that between 1980 and 2011, improvements in energy intensity amounted to only about 12% in ASEAN, far below the roughly 38% observed in OECD countries and 74% in China (Figure 1.1).

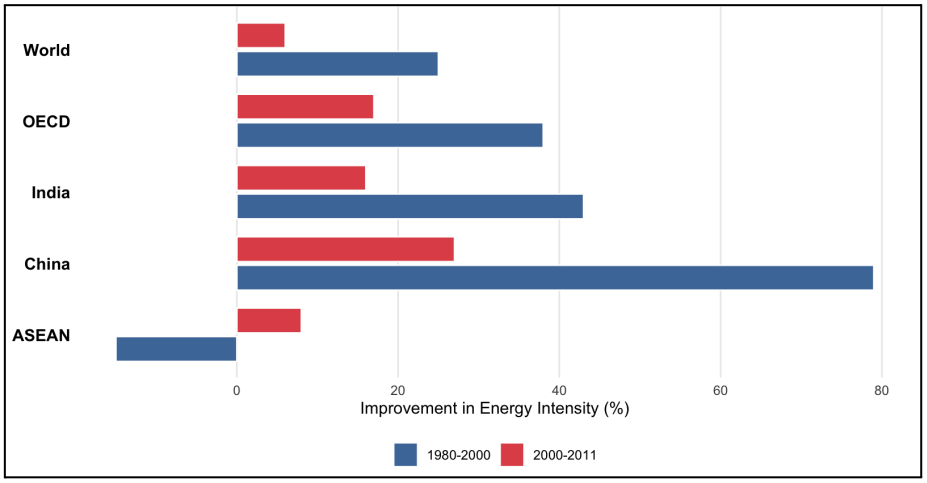

Figure 1.1 **Industrial energy intensity trends in selected countries and regions, 1980–2000 and 2000–2011**

Source: Southeast Asia Energy Outlook (2013)

This widening efficiency gap between frontier innovators and late-industrializers underscores a fundamental reality: the transition to a low-carbon industrial base is not an automatic outcome of economic growth. Rather, the slow pace of intensity reduction is a symptom of a deeper systemic deadlock. In practice, the development and deployment of low-carbon technologies is an effort hindered in practice by structural imbalances in global innovation and the presence of deep-seated market failures. Despite their apparent cost-effectiveness, energy-saving innovations are currently underinvested and remain underutilized. In the realm of low-carbon technology, progress faces a dual challenge by the 'Twin Market Failures' (Jaffe and Stavins, 1994; Jaffe et al., 2005). On one hand, the cost of negative externalities from pollution are often not borne by the polluters themselves. On the other hand, the development of low-carbon technologies requires a capital-intensive experimental process that generates positive spillovers benefiting parties beyond the original investors (Rodrik, 2014).

Studies have shown that research and development (R&D) in low-carbon technologies is still concentrated in a few top-innovator countries, such as the US, Japan, and Germany (Verdolini and Galeotti, 2010). However, if the deployment of

these technologies is crucial to break free from a carbon-locked production system, it becomes critical to extend the focus beyond frontier innovators to encompass developing nations and late industrializers. For developing nations, overcoming the dual market failures of innovation does not necessarily mandate reinventing the wheel. Instead, international trade, and specifically integration into Global Value Chains (GVCs), serves as a critical conduit for technology acquisition (Keller, 2004).

Standard neoclassical models traditionally framed technology as a freely available good on an 'international shelf' that could be costlessly assimilated (Lall, 1992). However, as Coe and Helpman (1995) prominently established, international technological progress is not acquired passively, but rather is predominantly transmitted through trade in intermediate goods. When a downstream manufacturing firm in a developing country imports advanced machinery or high-tech intermediate components from frontier innovators, it effectively 'imports' the R&D embedded within those goods (Schmookler, 1966; Griliches, 1979). This mechanism of embodied technology transfer allows late-industrializers to bypass the prohibitive costs of initial R&D, leveraging the vertical spillovers generated by their upstream suppliers. Nevertheless, capitalizing on this knowledge transfer is a highly demanding process. It strictly requires 'absorptive capacity', defined as the inherent ability of a firm or economy to recognize the value of external knowledge, assimilate it, and apply it to their own production processes (Cohen & Levinthal, 1990; Lall, 1992).

Nowhere is the dynamics of technology diffusion more visible than in Southeast Asia. Over the past few decades, the region has emerged as a key node within “Factory Asia,” characterized by dense cross-border production networks and extensive trade in intermediate goods (Baldwin, 2016). However, participation and technological capabilities within these networks remain highly heterogeneous. At one end of the spectrum, Singapore has developed highly sophisticated technological capabilities and a technologically advanced export structure, reflecting its status as a high-income economy with strong absorptive capacity (Lall, 2000). In contrast, many of the ASEAN-5 economies continue to rely heavily on processes of technological learning and capability accumulation characteristic of late-industrializing firms (Hobday, 1995; Intarakumnerd et al., 2002). Less-industrialized economies such as Cambodia, Laos, and Myanmar remain largely peripheral to these manufacturing networks due to

significant structural differences in industrial capacity and development levels, widely described in the literature as the ASEAN development divide (Hill and Menon, 2013). Within this context of technological learning and industrial upgrading, improvements in production efficiency become a critical dimension of economic transformation. As illustrated in Figure 2, the aggregate manufacturing energy intensity across these five nations has exhibited a discernible downward trend over recent decades.

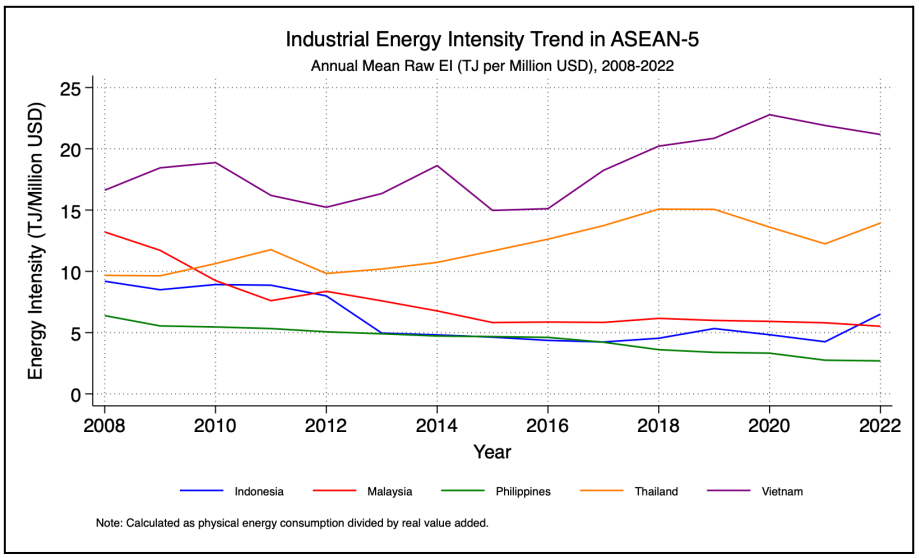

Figure 1.2 **Industrial energy intensity trend in ASEAN-5, 2008-2022**

Source: ADB MRIO and APEC EGEDA

This study seeks to fill that gap by not only analyzing the internal determinants of energy intensity, but also by exploring the vertical propagation of efficiency across industrial supply chains. To rigorously examine this phenomenon, this study isolates five emerging economies, Indonesia, Malaysia, the Philippines, Thailand, and Vietnam (ASEAN-5), as the primary analytical focus. The selection of ASEAN-5 provides an ideal empirical laboratory to examine vertical technology spillovers. These nations share a broadly homogenous stage of industrialization, functioning primarily as manufacturing assemblers heavily dependent on foreign technology inputs (Athukorala, 2011; Baldwin and Lopez-Gonzalez, 2015). Consequently, while their structural reliance on foreign inputs presents a theoretical pathway for green technology adoption, the actual extent to which vertical spillovers drive energy intensity reductions in these late-industrializing economies remains an unresolved empirical puzzle. Clarifying this

dynamic is therefore essential to determine whether global supply chain integration can effectively accelerate industrial decarbonization in the Global South.

1.2 Research Gap

The literature on energy intensity and energy efficiency has expanded rapidly. Studies in developing regions, including much of Asia, commonly rely on decomposition frameworks and spatial econometric methods to document energy intensity trends and their drivers, typically at the aggregate or sectoral level (Fisher-Vanden et al., 2004; Ma & Stern, 2008; Saraswati et al., 2022). Much of the developing-country literature continues to treat sectors as independent units and overlooks how energy efficiency improvements propagate across sectors (Azaliah et al., 2024). In advanced economies, however, the literature has moved beyond aggregate trends to explicitly incorporate sectoral dynamics, efficiency improvements, and technology-driven changes in energy use (Mulder & Groot, 2012; Wan et al., 2015; Costantini et al., 2017; Sun et al., 2021; Rahko, 2025).

Still, the evidence found in advanced economies cannot be directly applied to developing nations because innovation regimes differ fundamentally. While studies in advanced economies proxy improvement from technology using green patents, developing countries are largely technology adopters. Technological improvements largely arrive through imported capital goods, intermediate inputs, and foreign direct investment (Imbruno and Ketterer, 2016; Pane and Patunru, 2022), making trade-based measures more relevant than domestic patenting.

This study seeks to address these gaps by adapting the vertical spillover framework developed for advanced economies (Rahko, 2025) to ASEAN-5 and their main trading partners, explicitly accounting for the role of ASEAN economies as technology adopters.

1.3 Research Questions

Guided by the background and gaps outlined above, this study seeks to answer the following research questions:

1. How do changes in the energy intensity of upstream industries influence the energy intensity of downstream industries through input–output production linkages?

2. To what extent do imported intermediate inputs from upstream industries generate vertical technological spillovers that reduce the energy intensity of downstream industries?

Answering these questions will clarify the role of trade and supply-chain integration as actionable policy levers for accelerating industrial decarbonization beyond the limits of domestic innovation alone.

1.4 Research Objectives

Guided by the research questions outlined above, this study pursues three main objectives. First, it examines the causal impact of vertical energy intensity spillovers by investigating whether efficiency gains in upstream industries propagate through supply-chain linkages to reduce industrial energy intensity in the ASEAN-5 region. Second, it evaluates the role of embodied technology spillovers by assessing whether the direct intake of high-technology intermediate inputs from global technology leaders improves the energy performance of downstream manufacturing sectors in ASEAN-5. Finally, the study aims to provide empirical evidence on supply-chain integration as a potential policy lever for industrial decarbonization in late-industrializing economies, particularly in addressing the constraints created by domestic market failures in green innovation.

1.5 Research Contributions

This study provides four primary contributions to the literature on energy economics and international trade in developing regions. First, it adapts the measurement of technology diffusion to the specific context of late-industrializing economies. While existing studies in advanced economies (Rahko, 2025; Popp, 2001) primarily use patent data as a proxy for innovation, this research argues that for the ASEAN-5, technological upgrading is a process of absorption rather than invention. By constructing a targeted Imported Technology Spillover variable, this study captures the "embodied" knowledge found in high-tech capital goods. This provides a more accurate and relevant metric for measuring how developing nations leverage the global "technological shelf" to reduce industrial energy intensity.

Second, it contributes through the harmonization of high-resolution regional datasets that offer a granular view of Southeast Asian industrial performance. By

integrating the ADB Multi-Regional Input-Output (MRIO) tables with the APEC Energy Database, this research constructs a unique panel dataset that links economic transaction flows with physical energy consumption at a detailed sectoral level. While global databases such as WIOD often offer limited coverage or high aggregation for Southeast Asian economies, the use of these specialized regional datasets allows for a precise and context-specific analysis of the ASEAN-5. This data integration provides a robust empirical foundation that bridges the gap between trade statistics and industrial energy performance.

Third, it addresses the fundamental identification challenge in supply-chain research using a robust econometric strategy. A significant problem in vertical spillover studies is the endogeneity of supplier selection; downstream industries may selectively switch to efficient suppliers because they are already aiming for higher efficiency. To isolate the true causal effect of the newly constructed dataset, this study utilizes a Two-Stage Least Squares (2SLS) approach with a Shift-Share (Bartik) instrument. By fixing the supply-chain share to the 2007 base year, we isolate the exogenous variation driven by global technological shocks from the endogenous changes in supplier selection. This methodological rigor ensures that the results reflect the impact of upstream progress rather than mere strategic shifts in domestic procurement.

Lastly, it provides empirical insights into the dynamic production network of 'Factory Asia' and offers actionable policy implications. Industrial supply chains in the ASEAN-5 are characterized by high volatility, where suppliers often change due to global capacity shifts, geopolitical reconfigurations, or China’s rising dominance rather than domestic firm-level decisions alone (Antràs, 2020; Gereffi et al., 2021). By capturing these inter-sectoral interdependencies, this study moves beyond the traditional view of industries as isolated units. It provides actionable evidence for policymakers that trade integration is a critical lever for industrial decarbonization, proving that a region’s energy performance is a collective outcome of its production network.

CHAPTER 2
LITERATURE REVIEW

2.1 Theoretical Framework

This chapter reviews the theoretical foundations underlying vertical supply-chain linkages and their implications in shaping the energy intensity of ASEAN-5 industries. These concepts provide the analytical framework for understanding how knowledge and productivity is embodied in the inputs purchased from the supplier. The discussion first establishes the theory behind the construction of energy intensity as a key measure, before examining the theoretical mechanisms through which vertical spillovers and international technology diffusion occurs.

2.1.1 Energy Intensity

At a macroeconomic level, energy intensity is a fundamental metric representing the amount of energy consumed per unit of economic output, serving as a primary indicator of how efficiently an economy uses energy to generate goods and services (Thaler, 2011; Fiorito, 2013). Conceptually, the critical importance of energy intensity can be understood through standard decomposition frameworks of energy-related CO_2 emissions. As illustrated in Figure 2.1, the Kaya identity (Kaya, 1990) expresses aggregate emissions as the product of population, output per person, energy intensity, and carbon intensity. This framing highlights two broad and complementary pathways for global decarbonization: reducing the carbon content of the energy supply itself, and reducing the total energy required to produce economic output. While the former relies heavily on fuel switching and renewable deployment, the latter operates strictly through efficiency improvements, technological progress, and optimized production processes (Mulder & Groot, 2012).

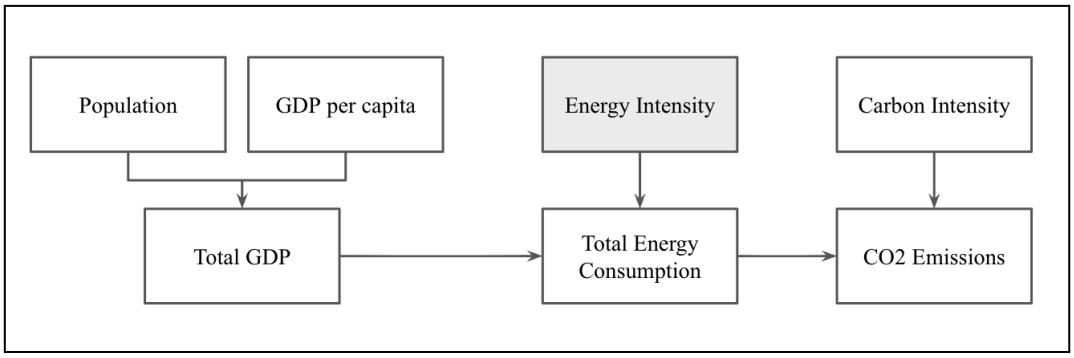

Figure 2.1 **Kaya Identity**

Source: Kaya, 1990

However, the interpretation of energy intensity is highly nuanced. Rather than reflecting pure technical efficiency alone, it is shaped by an economy's industrial structure, policy interventions, and sectoral shifts (Samuelson, 2014; Khatib, 1995; Radovanović and Filipović, 2015; Rahko, 2025). A declining aggregate energy intensity may result either from genuine efficiency improvements within specific industries or from structural shifts toward less energy-intensive sectors, such as services (Proskuryakova and Kovalev, 2015; Filippini and Hunt, 2015). However, empirical literature reveals that historical declines in energy intensity are predominantly explained by within-sector technological improvements, rather than structural shifts toward cleaner sectors (Wurlod and Noailly, 2018; Zhang, 2013). Wan et al. (2015) explicitly attribute this internal improvement to technological upgrading and innovation.

To understand the mechanics of this within-sector technological upgrading, the analysis must bridge microeconomic production behaviors with aggregate industrial outcomes. At the sectoral level, this production structure is formally captured by the KLEM (Capital, Labor, Energy, and Materials) framework, pioneered by Berndt and Wood (1975) and Hudson and Jorgenson (1974). Within this model, energy is recognized not merely as a byproduct, but as a distinct and irreplaceable primary factor of production. Consequently, energy intensity is not a static ratio but a dynamic outcome of the Marginal Rate of Technical Substitution (MRTS) between energy and other inputs, particularly capital (Apostolakis, 1990).

This dynamic nature underscores that a firm's optimal input mix is highly sensitive to prevailing economic signals (Pindyck, 1979). This sensitivity thus forms the theoretical core of the Induced Innovation Hypothesis (Hicks, 1932; Acemoglu, 2002),

which demonstrates how profit-maximizing firms systematically redirect investments toward augmenting the relatively scarcer and more expensive factors of production. When energy prices rise, firms are heavily incentivized to substitute energy with capital by adopting 'energy-saving' techniques to maintain competitive cost structures (Newell et al., 1998; Popp, 2002; Linn, 2008). The cumulative macroeconomic result of this micro-level, price-induced technological adjustment is a structural decline in the industry's aggregate energy requirement relative to its economic output (Sue Wing, 2008; Metcalf, 2008; Fisher-Vanden et al., 2004).

This dynamic is especially relevant for manufacturing-heavy emerging economies. Because their structural shifts away from industry are constrained by their established roles as global production hubs, achieving energy intensity reductions relies critically on these genuine within-industry efficiency improvements (Voigt et al., 2014). Therefore, understanding how these technological capabilities are formed necessitates an examination of how technology is externally acquired. In contexts where domestic innovation systems remain emerging, previous studies demonstrate that technological upgrading is predominantly driven by the mechanisms of vertical spillovers (Javorcik, 2004; Blalock and Gertler, 2008). This is particularly pronounced within capital-intensive manufacturing sectors, where strong vertical supply-chain linkages and the importation of embedded knowledge serve as the primary conduits for enhancing local industrial productivity and structural energy efficiency (Kuswardana et al., 2021).

2.1.2 Mechanisms of Vertical Spillovers

The idea of inter-sectoral spillovers can be traced back to the linkage theory by Hirschman (1958), which posits that developments and investments in one industry generate forward and backward linkages that can stimulate economic activities and capability development across related sectors. Within the technology diffusion literature, spillovers are commonly distinguished into horizontal (intra-industry) and vertical (inter-industry) spillovers (Keller, 2004; Görg and Greenaway, 2004). Marshall (1890) theorized that horizontal spillovers occur among firms operating within the same industry, often through mechanisms such as imitation or labor mobility. This theory is later formalized in the Marshall-Arrow-Romer framework (Arrow, 1962; Romer, 1986), which highlights how the geographic concentration of firms within the same industry facilitates knowledge spillovers and technological progress. However, such spillovers

are often limited by competitive incentives and firms' efforts to protect proprietary knowledge (Aitken and Harrison, 1999). On the other hand, the mechanism for vertical spillovers are connected through direct input–output relationships within supply chains, making knowledge transmitted between upstream suppliers and downstream buyers through production linkages, supplier development, and quality upgrading requirements (Javorcik, 2004; Keller, 2004).

The theoretical framework for vertical spillovers rest on the premise that innovation is not contained within the firm that generates it. Griliches (1979) distinguishes two mechanisms through which technological change may diffuse across firms and industries. The first one is rent spillovers, which arise through market transactions such as the purchase of technologically improved inputs, while the second one is pure knowledge spillovers, which occur when ideas or technological knowledge diffuse without direct economic transactions.

2.1.2.1 The Physical Channel: Embodied Technology and Imported Inputs

Through the first mechanism, the technological improved inputs are diffused through trade in intermediate goods (Grossman and Helpman, 1991). This perspective is consistent with the demand-driven view of innovation proposed by Schmookler (1966), who argued that inventive activity responds to market demand and investment opportunities. As downstream industries expand and demand more sophisticated inputs, upstream suppliers are incentivized to innovate in order to meet these evolving production requirements. Consequently, technological change in supplier industries may diffuse to user industries through the purchase and use of improved intermediate goods.

Early studies have heavily relied on patent citations to track knowledge flows (Jaffe et al., 1993). While such measures provide useful information on codified knowledge diffusion, they remain an imperfect proxy for technological spillovers. Patent citations may not fully capture the transfer of tacit or embodied knowledge that occurs through commercial transactions, such as the purchase and use of intermediate goods in production processes. As a result, they may be less suitable for analyzing technology diffusion in technology-adopting economies where learning frequently occurs through trade and production linkages (Huang et al., 2022).

2.1.2.2 The Non-Technological Channel: Intangible Knowledge and General Productivity

Beyond the physical embodiment of innovation, vertical spillovers are also driven by non-technological mechanisms. First, supply-use relationships between buyers and suppliers often require regular user-producer interactions. According to Lundvall (1988), innovation is not an isolated laboratory activity; it is an interactive learning process involving the exchange of tacit knowledge between producers and users. Through these interactions, firms may exchange technical know-how, operational experience, and feedback on product performance. In evolutionary theory, Nelson and Winter (1982) argue that firms accumulate knowledge in the form of organizational routines, defined as recurrent patterns of behavior that embody production knowledge and capabilities, which can shape how technologies and practices diffuse across economic relationships. Thus, when accounting for general productivity spillovers, it is important to account for the spillovers that can not be explained by patent or other technological measures.

It is also important to note that the effectiveness of vertical knowledge transmission is not automatic. Building on the notion of social capability by Abramovitz (1986) and the absorptive capacity framework developed by Cohen and Levinthal (1990), the successful internalization of international knowledge spillovers requires sufficient technological and organizational capacity within recipient industries. Firms must possess the ability to recognize, assimilate, and apply external knowledge in order to translate spillovers into productivity gains. Furthermore, the intensity of these vertical spillovers may be subject to distance decay, where the strength of knowledge flows declines as geographical or cultural distance increases, reflecting the localized nature of knowledge transmission (Jaffe, 1989; Audretsch and Feldman, 1996; Keller, 2002). These theoretical considerations highlight the importance of both industrial capabilities and proximity in shaping the transmission of efficiency gains across vertically linked industries.

2.2. Empirical Evidence

While the theoretical frameworks establish the underlying mechanisms of factor substitution and knowledge diffusion, empirical evidence on how these dynamics interact in practice remain fragmented. This study bridges two historically distinct

branches of empirical literature. The first branch explores the determinants of energy intensity, starting from economic, technological, until institutional and behavioral factors. The second branch examines industry-level productivity spillovers, specifically focusing on the empirical studies analyzing the mechanisms through which embodied technologies and intangible knowledge are transferred.

2.2.1 Determinants of Energy Intensity

The structural drivers of energy intensity are inherently multifaceted. Empirical literature broadly categorizes these determinants into three distinct groups: economic factors (Bekun et al., 2025; Miketa, 2001; Kazmaganbetova et al., 2016; Shakya et al., 2022; Solarin, 2020), technological advancements (Ajayi & Reiner, 2020; Huang et al., 2022; Huang et al., 2017; Huang & Yu, 2016), and institutional or behavioral influences (Long et al., 2023; Alvarado et al., 2023; Yang et al., 2024; Azhgaliyeva et al., 2020; Verbič et al., 2017; Sun et al., 2019).

At the sectoral and operational level, input prices and production efficiency act as primary catalysts for change (Popp, 2002; Greening et al., 2000). Empirical studies done by Ajayi and Reiner (2020), Wurlod and Noailly (2018), Zhang (2013) similarly highlighted that higher energy prices increases production costs, which in turn compels firms to adopt energy-saving technologies and robustly reduces energy intensity. On a similar level, the depth of value creation within the production process also plays a pivotal role. Ajayi and Reiner (2020) found that as industries expand their gross value added using the same or improved level of inputs, they generate economies of scale. This condition translates directly to enhanced aggregate efficiency and a subsequent decline in industrial energy intensity.

Beyond the sectoral level, a nation's integration into the global economy through trade exerts a complex, dual effect on energy intensity (Wan et al., 2015). Although trade openness can incentivize domestic industries to upgrade their energy productivity due to heighten market competition and international knowledge diffusion (Wan et al., 2015), it can also potentially increase aggregate energy intensity if a country’s comparative advantage leads to structural specialization in highly energy-intensive and carbon-heavy export manufacturing (Chintrakarn, 2013; Huang et al., 2017). Ultimately, these sectoral and trade dynamics culminate in broader macroeconomic trends.

Previous studies in this field have confirmed that energy intensity tends to rise with economic growth up to a certain point, then declines as economies mature and shift toward less energy-intensive sectors (Bekun et al., 2025; Kazmaganbetova et al., 2016; Petrović et al., 2018). Huang and Yu (2016) attributed this inverted U-shaped relationship to the process of industrialization which expands cross-sectoral growth, labor demand, and energy input. However, as the economy grows, the growing affluence in the service sector development triggers a technique effect where economies invest in better technology and transition to services, which yield more value-added per unit of energy (Solarin, 2020). Petrović et al. (2018) uncovered a similar finding in Europe that higher GDP per capita is driving societal demand for a cleaner environment and stricter regulations, forcing companies to adopt energy-saving technologies.

However, while these macro-level structural shifts undeniably play a part in reducing aggregate energy intensity, studies on Eastern Europe and Central Asia shows that convergence in energy intensity is highly attributed to efficiency improvements rather than mere structural shifts (Zhang, 2013; Voigt et al., 2014). This is an important factor to consider when examining the efforts of developing countries to catch up with their predecessors. In the extensive literature of energy intensity for developed nations, this efficiency improvement is ascribed to the role of green innovation (Huang et al., 2022; Huang & Yu, 2016), which is particularly relevant as a contributor in energy-intensive sectors (Wurlod and Noailly, 2018; Ajayi and Reiner, 2020; Popp, 2001). Here, green innovation is quantified by total number of patents (Bottazzi and Peri, 2003; Brunnermeier and Cohen, 2003; Castaldi et al., 2015; Costantini et al., 2017).

Of course, these innovations and efficiency improvements do not come out of thin air. Although significant, energy-saving benefits of non-technological factors have received somewhat less attention in the literature (Rahko, 2025). Policy instruments and institutional quality influences the degree of improvement (Sun et al., 2019; Kozhukhova et al., 2019; Apergis and Garćıa, 2019). For instance, behavioral factors such as good management and energy management practices are argued to reduce industrial energy use (Backlund et al., 2012; Johansson and Thollander, 2018). Relatedly, Martin et al. (2012) show that climate-friendly management practices are strongly associated with higher energy efficiency and productivity in firms. Thus,

Rahko (2025) emphasizes the importance of accounting for broader productivity and efficiency improvements that are not captured by patent-based indicators.

The distinction between patent-based innovation and broader productivity improvements is particularly relevant when examining developing economies. Unlike advanced economies operating at the technological frontier, late-industrializing countries primarily grow through the adoption, adaptation, and learning of technologies developed elsewhere (Amsden, 2001). In this context, industrial upgrading and reductions in energy intensity are less associated with the limited domestic patenting activity and are instead strongly influenced by external technological diffusion. A large body of literature attributes technological progress in late-industrializing economies to openness to foreign direct investment (FDI) and the import of technologically advanced capital and intermediate goods, through which knowledge is transferred in embodied form (Huang et al., 2017; Pane and Patunru, 2022). However, to fully understand how these external shocks effectively lower energy intensity, the analysis must move beyond aggregate national trends and examine the inter-industry transmission and spillovers of knowledge.

2.2.2 Industry-level Productivity Spillover Effects

Building on the recognition that technological progress often originates outside the firm or sector itself, a substantial body of empirical literature examines how productivity improvements diffuse across industries through inter-sectoral linkages and production networks (Terleckyj, 1974; Scherer, 1982). Subsequent empirical studies further expanded this insight by examining how knowledge flows transmitted through production networks influence productivity outcomes. Delera and Foster-McGregor (2023) found that domestic industry-level total factor production (TFP) is highly elastic to knowledge flows imported through intermediate trade in GVCs. They also found that vertical spillover is particularly beneficial for countries further from the technology frontier. Across OECD countries, Serrano-Domingo and Cabrer-Borrás (2017) found that direct spillovers generated by knowledge produced by a sector's input trade partners significantly increase the sector's labor productivity, while indirect spillovers from other non-directly trade-related sectors do not significantly affect sector productivity.

Beyond general productivity improvements, a growing body of literature examines spillovers related specifically to energy efficiency and environmental

productivity. Wan et al. (2015) found that international trade flows explain 30% to 40% of the unobserved variation in positive energy productivity spillovers. They stated that sectors and countries with higher dependence on trade experience both higher energy productivity growth and a faster rate of efficiency convergence. Similarly, a study done in European industries by Constantini et al. (2017) found that in reducing emission intensity, the effect of foreign clean technologies incorporated into inputs acquired through global supply chains is even stronger than domestic upstream spillovers. Building on that finding, Rahko (2025) found that a 10% decrease in the energy intensity of upstream sectors results in a 2.4% to 4.6% reduction in the energy intensity of downstream sectors.

These internal operational shifts are further shaped by supply chain linkages, particularly the integration of service inputs (Rahko, 2025). Liang et al. (2025) found evidence from China that the liberalization and increased utilization of upstream productive services serve as vital conduits for intangible knowledge spillovers to downstream manufacturing. However, a significant theoretical contradiction exists regarding the relative magnitude of these vertical spillovers compared to direct domestic innovation. A study done by Costantini et al. (2017) assert that inter-sectoral linkages and spillovers are highly critical, often exerting a stronger influence on environmental efficiency than a sector's direct internal innovations. Conversely, research within different geographical contexts, such as Italy, suggests that while spillovers do improve environmental efficiency, their overall effect is significantly weaker than direct investments in green technology (Ghisetti and Quatraro, 2017; Jiao et al., 2020).

The effectiveness of these spillovers may also depend on spatial proximity and the nature of inter-industry relationships. Although international intermediate goods trade might be an ineffective channel for knowledge transfer, Serrano-Domingo and Cabrer-Borrás (2017) found evidence that productivity is significantly enhanced by direct vertical spillovers from a sector's domestic trade partners. Similarly, Delera and Foster-McGregor (2023) found that geographical distance does dampen vertical R&D spillovers. Rahko (2025) found that only the domestic spillovers yielded a significant effect in vertical spillovers from patented green technologies, suggesting that geographic proximity plays a critical role in innovation diffusion through the supply chain. Consequently, a downstream firm may improve its energy efficiency simply by

adopting a new organizational practice or transnational environmental standard acquired through its close business relationship with an innovative upstream supplier (Costantini et al., 2017; Wan et al., 2015).

Many studies in developed countries rely on patents and R&D expenditures to measure innovation. For instance, Wurlod and Noailly (2018) argued that patents filed simultaneously at European, Japanese, and US offices reliably filter out low-value inventions, thus serving as an excellent indicator of highly valuable technological advancements. Constantini et al. (2017) also justified the use of patents by stating that patents represent a close, flexible, and readily available proxy for green product innovations. However, numerous studies explicitly outline why R&D and patents are imperfect. Verdolini and Galeotti (2011) note that patent data only sheds light on embodied technological change (innovations embedded in new machinery). Rahko (2025) added that patents completely miss non-technological innovations, such as climate-friendly management practices, energy audits, and behavioral and organizational changes.

In the study of developing countries, Evenson and Westphal (1995) and Hausmann and Rodrik (2003) argue that traditional R&D and patent metrics completely fail because much crucial technology is tacit and cannot be codified into blueprints. Unlike advanced economies operating at the technological frontier, late-industrializing countries primarily grow through the adoption, adaptation, and learning of technologies developed elsewhere (Amsden, 2001). In this context, industrial upgrading and reductions in energy intensity are less associated with the limited domestic patenting activity and are instead strongly influenced by external technological diffusion. A large body of literature attributes technological progress in late-industrializing economies to openness to foreign direct investment (FDI) and the import of technologically advanced capital and intermediate goods, through which knowledge is transferred in embodied form (Huang et al., 2017; Pane and Patunru, 2022).

2.3. Conceptual Framework

To better understand the intuition behind this study, a conceptual framework is presented in Figure 2.1. In this study, we aim to analyze how vertical spillovers from global upstream suppliers impact industrial energy intensity across the ASEAN-5 region. The paper's intuition is based on the theories of embodied technology and

inter-industry knowledge diffusion, which posit that technological progress in upstream industries is transmitted downward through intermediate inputs (Schmookler, 1966; Griliches, 1979; Wolff & Nadiri, 1993). While domestic innovation in green technology often suffers from market failures due to high upfront costs and uncertain returns (Jaffe et al., 2005; Rodrik, 2014), these vertical linkages offer a compensatory mechanism for late-industrializing economies to 'absorb' efficiency gains from the global technological frontier (Lall, 1992).

In order to identify this causal relationship, we adapt the framework of Rahko (2025) and Goldsmith-Pinkham et al. (2020) by utilizing Shift-Share (Bartik) instruments based on 2007 base-year supply chain weights. This approach ensures that the identified impacts reflect exogenous shocks from global suppliers rather than endogenous procurement strategies. Specifically, we distinguish between energy efficiency spillovers (process-based learning) and imported technology spillovers (direct capital-based adoption), as these two channels represent distinct pathways for industrial upgrading. Furthermore, we control for other factors that influence sectoral energy intensity, including: relative energy prices to capture price-induced substitution (Hicks, 1932; Popp, 2002), value-added productivity (Berndt & Wood, 1975; Voigt et al., 2014), and overall trade and service integration (Wan et al., 2015; Costantini et al., 2017; Setyawan, 2020).

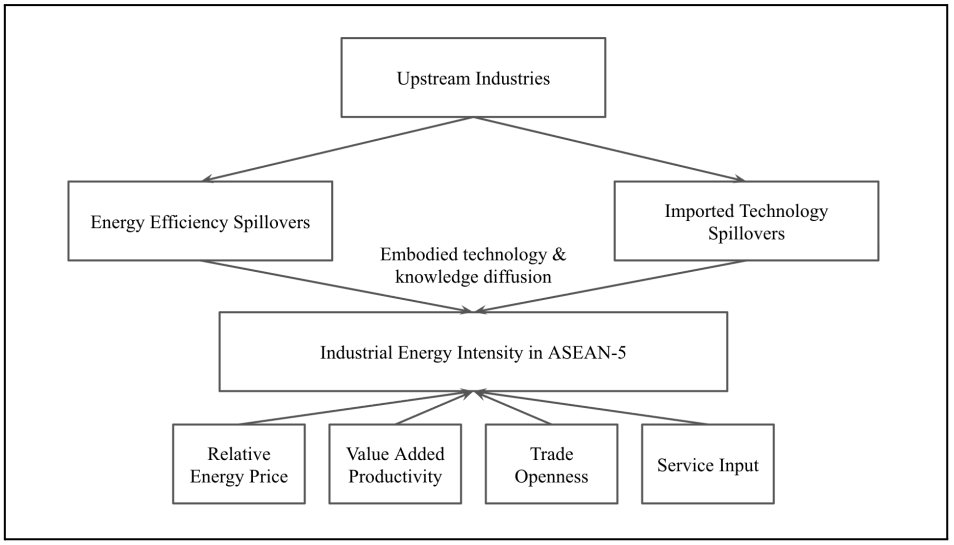

Figure 2.2 **Conceptual Framework**

CHAPTER 3
METHODOLOGY

3.1 Data and Variables

3.1.1 Data Source and Coverage

To analyze the vertical technology and energy productivity spillovers along the ASEAN-5 industrial supply chain, this study employs two complementary datasets. First, the study utilizes the Asian Development Bank Multi-Regional Input-Output (ADB MRIO) tables for 62 economies, available for year 2000 and year 2007–2024. The nominal dataset is used to construct coefficients that reflect actual market transactions and relative trade shares in a given year, while the 2010 constant price tables are used to extract real value-added figures and used as sectoral deflators. Both the current prices (nominal) and constant 2010 prices (real) versions of these tables are used to ensure that while transaction patterns are captured in their current market context, the measurement of productivity or intensity remains insulated from the distortions of inflation and sectoral price volatility.

The ADB MRIO dataset maps inter-industry transactions across 62 economies and 35 sectors. However, aligning with the research objective to analyze late-industrializing manufacturing hubs, the final subject of analysis, which is the downstream buyers evaluated for their energy intensity (the dependent variable), is strictly narrowed down to 13 core manufacturing sectors. This specific sample consists of sectors c3 through c7 and c9 through c16. Other industrial sectors such as Mining (c2), Coke and Refined Petroleum (c8), Electricity, Gas, and Water Supply (c17), and Construction (c18) are systematically excluded from the subject list. Specifically, c8 and c17 are excluded because they act predominantly as upstream energy producers rather than standard industrial consumers. However, it is crucial to note that while these sectors are not the subjects of the analysis, the full 35-sector matrix (including c8 and c17) is continuously utilized as the supplier poo' to calculate the structural transaction weights and the upstream spillovers.

The second dataset utilized is the Asia-Pacific Economic Cooperation Energy Database (APEC EGEDA) Energy Balance Table, which is used for its sectoral energy consumption data. This dataset is available from 1980–2022. The decision to use the

APEC EGEDA dataset is driven by its high degree of regional specificity and its detailed reporting of sectoral energy balances. Critically, the database follows a classification system rooted in the International Standard Industrial Classification (ISIC), which is identical to the framework used by the ADB MRIO. This structural alignment is essential for the study as it facilitates a precise sectoral concordance and mapping process. The analysis using the total final energy for the industrial block, which consists of 13 sectors, will be utilized for the calculation of energy intensity and relative energy price.

3.1.2 Empirical Scope and Sample Selection

The empirical scope of this study is categorized into two distinct groups of actors, the subjects of analysis and the sources of spillovers. The outcome variable, the downstream industrial energy intensity, is exclusively focused on the ASEAN-5 economies consisting of Indonesia, Malaysia, the Philippines, Thailand, and Vietnam. This regional selection provides a homogeneous sample of late-industrializing manufacturing hubs that share similar developmental trajectories (Athukorala, 2011; Baldwin and Lopez-Gonzalez, 2015).

Spillovers from upstream energy intensity are calculated using source sectors from APEC member economies where the data for both physical energy consumption from APEC EGEDA and economic value-added data from ADB MRIO are available. Following the logic of Rahko (2025), the 'Rest of the World' (RoW) group and non-APEC countries are excluded from the efficiency shock calculation because sectoral energy intensity cannot be computed for these aggregated or data-limited regions.

For the calculation of imported technology spillovers, this study identifies a specific subset of nine 'tech leaders' within the APEC pool as the primary sources of embodied technology, namely the United States, Japan, China, South Korea, Canada, Australia, Mexico, Chinese Taipei, and Singapore. The selection of these specific economies is driven by two methodological rationales. First, restricting the source countries to the existing APEC pool ensures structural consistency with the baseline energy spillover calculations. Second, these nine economies represent the technological frontier within the Asia-Pacific global value chain. They possess a distinct comparative advantage in aggregate R&D investments and dominate the export of advanced capital

goods and machinery. By deliberately filtering the source pool to these specific tech-abundant economies, the model ensures that the calculated spillovers genuinely capture the intake of sophisticated frontier technologies rather than standard low-tech intermediate inputs. Furthermore, spillovers in this category are only captured from R&D-intensive sectors based on the OECD Taxonomy of Economic Activities Based on R&D Intensity (2016). The sectors include Chemicals (c9), Basic Metals (c12), Machinery (c13), Electrical and Optical Equipment (c14), and Transport Equipment (c15).

3.1.3 Sectoral Harmonization and Data Construction

A significant methodological challenge in merging the ADB MRIO and APEC EGEDA datasets is the difference in data granularity and sectoral classification structures. The APEC Energy Balance Table categorizes energy flows into three primary stages: Primary Energy, Transformation, and Final Energy consumption. To facilitate a precise cross-database mapping with the ADB MRIO's 35 economic sectors, this study employs a targeted, structure-aware allocation approach.

For standard manufacturing sectors (ADB sectors c3 through c16), energy consumption is fundamentally an end-use activity. Therefore, their data is extracted from the 'Final Energy' (Industrial) section of the APEC database. Because EGEDA reports industrial energy consumption only at an aggregated level, sectoral energy use must be approximated using proportional output shares derived from the ADB MRIO tables. This approach assumes that energy consumption scales with the production size of each sector, consistent with the concept of energy intensity defined as energy use per unit of economic output (United Nations, 2001). Similar approximations are commonly adopted in environmental input–output analysis, where sector-level averages are used when detailed process-level data are unavailable (Hendrickson et al., 2006).

Conversely, a different treatment is required for the upstream energy-producing sectors, namely Coke and Refined Petroleum (c8) and Electricity, Gas, and Water Supply (c17). While these two sectors are excluded from being the downstream subjects of the analysis, they are fundamental upstream suppliers. Their efficiency improvements heavily spill over into the manufacturing block, meaning their specific energy intensity must still be accurately computed to construct the upstream spillover variable (S_{ijt}^{EI}).

Because these sectors process and convert primary fuels into secondary energy products, they do not act as end-users. Consequently, their energy usage is recorded under the 'Transformation' section of the APEC EGEDA rather than the 'Final Energy' section. To prevent skewed calculations, c8 and c17 bypass the standard output-weighting allocation. Instead, their energy consumption is matched directly to their corresponding specific transformation line items in the APEC database ('Oil Refineries' and 'Electricity Plants').

Once the physical energy consumption is successfully mapped and estimated for each specific ADB sector using these distinct categorization logic, the final Energy Intensity is calculated by dividing this sectoral energy consumption by the real value added (extracted from the 2010 constant price ADB tables). This harmonized dataset serves as the foundation for the baseline regression and the construction of the instrumental variables.

Beyond the primary variables of interest, the construction of the relative energy price control variable ($\frac{P^E}{P^Q}$) requires an integration of both physical and monetary datasets. First, the total sectoral energy expenditure is quantified by aggregating the intermediate consumption values sourced from the ADB MRIO current price (nominal) tables. Specifically, this captures the total monetary purchases (in USD) made by each downstream ASEAN-5 manufacturing sector from all primary and secondary energy-producing sectors globally, namely, Mining and Quarrying (c2), Coke and Refined Petroleum (c8), and Electricity, Gas, and Water Supply (c17), across all 62 available source economies.

Once the total energy expenditure is aggregated, the nominal unit price of energy (P^E) is calculated by dividing this monetary expenditure by the total physical energy consumption estimated previously from the APEC EGEDA database. Next, to account for industry-specific inflation and varying economic valuations, the sectoral output deflator (P^Q) is derived by dividing the nominal total output by the real total output using the respective ADB MRIO tables. Finally, the relative energy price is obtained by taking the natural logarithm of the ratio between the nominal energy price and the sectoral deflator ($\frac{P^E}{P^Q}$). This rigorous construction isolates pure price signals, ensuring

that the econometric model accurately controls for price-induced energy efficiency improvements.

To conclude, a comprehensive overview of the data sources, measurements, and definitions for each variable is presented in Table 3.1.

Table 3.1 Variable Definitions and Data Sources

Dependent		
Variable	Operational Definition & Measurement	Source
Energy intensity $ln(EI_{ijt})$	The natural logarithm of sectoral energy intensity. Calculated as the ratio of an industry's total physical energy consumption to its real value-added (constant 2010 prices).	APEC EGEDA (Numerator), ADB MRIO (Denominator)

Independent		
Variables of Interest		
Variable	Operational Definition & Measurement	Source
Vertical Energy Spillover (S_{ijt}^{EI})	The weighted average of the natural logarithm of energy intensity across all upstream supplier industries. The weights are based on the share of intermediate input transactions, strictly excluding intra-industry purchases.	APEC EGEDA (Numerator), ADB MRIO (Denominator)
Spillovers from imported technology ($S_{ijt}^{ImpTech}$)	The weighted intake of embodied technology. Calculated as the total share of high-tech intermediate inputs sourced from 9 global tech leaders relative to total intermediate consumption.	ADB MRIO

Control Variable		
Variable	Operational Definition & Measurement	Source
Relative Energy Price ($\frac{P^E}{P^Q}$)	The natural logarithm of the relative energy price. Calculated as the nominal unit price of energy (P^E) divided by the sectoral output deflator (P^Q). Controls for price-induced energy efficiency improvements.	ADB MRIO, APEC EGEDA
Import Share (*ImpSh*)	The proportion of intermediate inputs sourced internationally (where the supplier country differs from the buyer country) out of total intermediate consumption. Controls for the degree of global market integration.	ADB MRIO
Service Share (*ServSh*)	The proportion of intermediate inputs sourced from tertiary service sectors (c19-c35) out of total intermediate consumption. Controls for the "servicification" trend in manufacturing.	ADB MRIO
Value added per output (*VA/O*)	The ratio of real value-added to real total output. Controls for an industry's structural production efficiency and its relative position within the value chain.	ADB MRIO

3.2 Model Specification

This study employs a two-pronged estimation strategy to examine the relationship between vertical spillovers and industrial energy intensity. First, this study utilized a Fixed Effects (FE) panel data model with the baseline estimating equation defined as follow:

$$ln(EI_{ijt}) = \beta_0 + \beta_1 S_{ijt}^{EI} + \beta_2 S_{ijt}^{ImpTech} + \delta X_{ijt} + \alpha_{ij} + \gamma_t + \varepsilon_{ijt} \quad (1)$$

Here, the dependent variable $ln(EI_{ijt})$ represents the natural logarithm of energy intensity for industry *i* in country *j* at time *t*, spanning the panel period from 2008 to 2022. By using the logarithmic form, the model allows for an interpretation of the coefficients as elasticities, meaning they reflect the percentage change in downstream energy intensity resulting from a percentage change in the spillover variables.

The core of the analysis lies in the two independent variables of interest, S_{ijt}^{EI} and $S_{ijt}^{ImpTech}$. The input-output tables from the ADB MRIO are utilized to capture supply-use relationships across industries and countries, following a similar approach to that adopted by Costantini et al. (2017) using the WIOD and Jiao et al. (2020) using China's input-output tables. The spillover variables are thus defined with the equation below:

$$S_{ijt}^{EI} = \sum_{c=1}^{C} \sum_{k=1}^{K} w_{ijck,t} \; lnEI_{ckt}, \; , \forall kc \neq ij \quad (2)$$

$$S_{ijt}^{ImpTech} = \sum_{c=1}^{C} \sum_{k=1}^{K} w_{ijck,t} \; ImpTech_{ckt}, \; , \forall kc \neq ij \quad (3)$$

Equation (2) represents vertical spillovers arising from overall upstream improvement in energy intensity, while Equation (3) captures the intake of embodied technology through intermediate imports. Using upstream energy intensity as a proxy for general vertical spillovers follows the methodological precedents established by Wan et al. (2015), Serrano-Domingo and Cabrer-Borrás (2017), and Rahko (2025), who all utilized similar measures to track the propagation of energy and labor productivity shocks along the supply chain. In this study, S_{ijt}^{EI} represents a 'process-based' spillover, capturing how general efficiency improvements in supplier industries, whether through organizational learning or incremental process changes, leak into the production function of downstream ASEAN-5 buyers.

By including both upstream energy intensity and high-tech imported inputs, the present analysis is able to disentangle the dual nature of vertical spillovers. It allows the study to uncover not just whether vertical spillovers exist, but also to separate between

two distinct causal mechanisms. The first one is through the embodied efficiency effect, where industries become cleaner simply by using more efficient inputs from any observed supplier, as also proved by past studies done by Wolff (2011), Badinger and Egger (2016), and Serrano-Domingo and Cabrer-Borrás (2017). Next, the second one is through the direct technology intake effect, where industries improve through the specific adoption of sophisticated machinery from the global technological frontier (Huang et al.; 2017; Huang & Yu, 2016; Delera & Foster-McGregor, 2023).

The weights (*w*) in Equation (2) and (3) are based on the supply chain linkages, mathematically written as follows:

$$w_{ijck,t} = \frac{m_{ijck,t}}{\sum_{c=1}^{C}\sum_{k=1}^{K} m_{ijck,t}}, \forall kc \neq ij \quad (4)$$

Where $m_{ijck,t}$ denotes the monetary value of intermediate inputs that industry *i* in country *j* purchases from sector *c* in country *k* at time *t*. The denominator represents the total intermediate consumption of industry *i*, effectively normalizing the weight such that $\sum w=1$ with the exclusion of the industry's own sector. The higher the share of inputs sourced from a specific supplier, the larger the potential for input-embedded spillovers. As specified in the condition $\forall kc \neq ij$, all intra-industry transactions are excluded to isolate purely vertical transmissions and prevent the reflection problem.

Other than the variables of interest, the estimating Equation (1) also incorporates control variables X_{ijt} consisting of relative energy price, import share, service share, and value added per output (Table 3.1) and a two-way panel fixed-effects estimation method to rigorously address potential endogeneity and omitted variable bias. First, by including industry-country fixed effects α_{ij}, the model effectively removes potential bias caused by unobservable characteristics unique to specific sectors within specific countries that do not change over time. This ensures that the estimates are not distorted by permanent structural factors, such as a country's inherent geographical endowments, baseline infrastructure, or deeply entrenched institutional and business practices within a specific domestic sector. Furthermore, the model employs year fixed effects γ_t to control for aggregate global shocks and specific time trends that simultaneously affect all industries across the ASEAN-5 region. This guarantees that the estimated spillovers are driven by actual supply chain dynamics rather than external macroeconomic events

in a given year, such as global financial crises, international supply chain disruptions, or sudden volatility in global energy markets (Rahko, 2025).

While the two-way fixed-effects estimation effectively mitigates biases from time-invariant unobservables and aggregate macroeconomic shocks, there are still persisting potential endogeneity surrounding the spillover variables. Rahko (2025) noted that in the context of inter-industry linkages, reverse causality is generally less pronounced than endogeneity issues related to an industry's direct internal innovations. This is because it is difficult for a downstream buyer to dictate the fundamental technological progress of its upstream suppliers, so the innovation happening upstream is largely considered exogenous (an external shock) to the buyer. However, a persistent econometric concern is that broader industry trends influencing downstream energy intensity might simultaneously alter the supply-use linkages or that the observed spillovers merely reflect joint sectoral trends across the ASEAN-5 manufacturing network.

To address these remaining endogeneity concerns, the model is further estimated using an instrumental variable (IV) approach. This instrumentation strategy follows the shift-share, or Bartik, instrumental variable approach (Bartik, 1991). The underlying rationale of this instrumentation is to utilize predetermined supply-use linkages as fixed weights that describe how different downstream industries are exposed to specific technological and energy shocks flowing from upstream suppliers. Recent theoretical advancements by Goldsmith-Pinkham et al. (2020) demonstrate that shift-share instruments provide a valid identification of causal effects provided that either the initial exposure weights or the upstream shocks are exogenous. By substituting contemporary transaction weights with pre-sample weights using the base-year linkages from 2007, the instrument effectively strips away any confounding variations that might endogenously alter supply chain configurations over the panel period. Consequently, this ensures that the estimated impacts on energy intensity reflect genuine vertical transmissions rather than concurrent structural adjustments.

To empirically implement this Bartik instrumental variable strategy, the model is estimated using a Two-Stage Least Squares (2SLS) panel regression framework. Formally, the instrumental variables for energy spillovers ($IV_{S_{ijt}^{EI}}$) and imported

technology spillovers ($IV_{S_{ijt}^{ImpTech}}$) are constructed by replacing the contemporary transaction weights with the static 2007 baseline weights as follows:

$$IV_{S_{ijt}^{EI}} = \sum_{c=1}^{C}\sum_{k=1}^{K} w_{ijck,\,2007}\ lnEI_{ckt'}\ ,\forall kc \neq ij \tag{5}$$

$$IV_{S_{ijt}^{ImpTech}} = \sum_{c=1}^{C}\sum_{k=1}^{K} w_{ijck,\,2007}\ ImpTech_{ckt'}\ ,\forall kc \neq ij \tag{6}$$

Where $w_{ijck,\,2007}$ represents the fixed intermediate input dependency of buyer industry i on supplier industry c in the pre-sample year 2007. Crucially, mirroring the conditions in Equations (2) and (3), intra-industry transactions are strictly excluded to satisfy the exclusion restriction and eliminate mechanical correlation.

Within the 2SLS framework, the first stage isolates the exogenous variation of the potentially endogenous spillover variables by regressing them onto these excluded Bartik instruments alongside all exogenous control variables and fixed effects. Assuming a full dual-IV specification where both spillovers are treated as endogenous, the first-stage equations are specified as follows:

$$S_{ijt}^{EI} = \pi_1 + \pi_{11} IV_{S_{ijt}^{EI}} + \pi_{12} IV_{S_{ijt}^{ImpTech}} + \pi_{13} X_{ijt} + \alpha_{ij} + \gamma_t + u_{ijt} \tag{7}$$

$$S_{ijt}^{ImpTech} = \pi_2 + \pi_{21} IV_{S_{ijt}^{EI}} + \pi_{22} IV_{S_{ijt}^{ImpTech}} + \pi_{23} X_{ijt} + \alpha_{ij} + \gamma_t + v_{ijt} \tag{8}$$

In these first-stage regressions, u_{ijt} and v_{ijt} are the idiosyncratic error terms. A highly significant π coefficient accompanied by a strong F-statistic indicates that the exogenous base-year linkage instrument is a powerful predictor of the contemporary spillover. The predicted values from these equations, denoted as $\hat{S}_{ijt}^{EI}$ and $\hat{S}_{ijt}^{ImpTech}$, represent the instrumented spillovers that have been systematically purged of endogeneity bias.

In the second stage, these fitted values are substituted back into the main structural equation to estimate the unbiased causal impact on industrial energy intensity:

$$ln(EI_{ijt}) = \beta_0 + \beta_1 \hat{S}_{ijt}^{EI} + \beta_2 \hat{S}_{ijt}^{ImpTech} + \delta X_{ijt} + \alpha_{ij} + \gamma_t + \varepsilon_{ijt} \tag{9}$$

Finally, to ensure the statistical validity of the 2SLS estimation, a series of standard post-estimation diagnostics will be conducted. Specifically, the Durbin-Wu-Hausman test will be utilized to empirically verify the endogeneity of the spillover variables. Furthermore, the relevance and strength of the Bartik instruments will be formally validated in the first stage using the Kleibergen–Paap Wald rk F-statistic to rule out weak instrument bias. Following the conventional rule-of-thumb in the instrumental variable literature, an F-statistic greater than 10 indicates that the instruments are sufficiently strong and that weak instrument bias is unlikely to be a major concern (Staiger and Stock, 1997; Stock and Yogo, 2005). Throughout all estimations, standard errors will be clustered at the industry-country level to robustly account for potential serial correlation and heteroskedasticity within the panels over time.

References

A. Greening, L., Greene, D. L., & Difiglio, C. (2000). Energy efficiency and consumption — the rebound effect — a survey. *Energy Policy*, *28*(6-7), 389–401. https://doi.org/10.1016/s0301-4215(00)00021-5

Abramovitz, M. (1986). Catching Up, Forging Ahead, and Falling Behind. *The Journal of Economic History*, *46*(2), 385–406.

Acemoglu, D. (2002). Directed Technical Change. *Review of Economic Studies*, *69*(4), 781–809. https://doi.org/10.1111/1467-937x.00226

Acemoglu, D., Aghion, P., Bursztyn, L., & Hemous, D. (2012). The Environment and Directed Technical Change. *SSRN Electronic Journal*, *102*(1). https://doi.org/10.2139/ssrn.1668575

Aitken, B. J., & Harrison, A. E. (1999). Do Domestic Firms Benefit from Direct Foreign Investment? Evidence from Venezuela. *American Economic Review*, *89*(3), 605–618.

Ajayi, V., & Reiner, D. (2020). European Industrial Energy Intensity: Innovation, Environmental Regulation, and Price Effects. *The Energy Journal*, *41*(4). https://doi.org/10.5547/01956574.41.4.vaja

Allen, R. C. (2009). *The British Industrial Revolution in Global Perspective*. Cambridge University Press.

Alvarado, R., Ortiz, C., Cuesta, L., & Tillaguango, B. (2022). Spillovers impact of institutional and economic factors in energy intensity. *Sustainable Development*. https://doi.org/10.1002/sd.2485

Amsden, A. (2001). *The Rise of "The Rest."* https://doi.org/10.1093/0195139690.001.0001

Antràs, P., & Gortari, A. (2020). On the Geography of Global Value Chains. *Econometrica*, *88*(4), 1553–1598. https://doi.org/10.3982/ecta15362

Apergis, N., & Garćıa, C. (2019). Environmentalism in the EU-28 context: the impact of governance quality on environmental energy efficiency. *Environmental Science and Pollution Research*, *26*(36), 37012–37025. https://doi.org/10.1007/s11356-019-06600-1

Apostolakis, B. E. (1990). Energy—capital substitutability/ complementarity. *Energy Economics*, *12*(1), 48–58. https://doi.org/10.1016/0140-9883(90)90007-3

Arrow, K. J. (1962). The Economic Implications of Learning by Doing. *The Review of Economic Studies*, *29*(3), 155–173. https://doi.org/10.2307/2295952

Athukorala, P. (2011). Production Networks and Trade Patterns in East Asia: Regionalization or Globalization? *Asian Economic Papers*, *10*(1), 65–95. https://doi.org/10.1162/asep_a_00045

Azhgaliyeva, D., Liu, Y., & Liddle, B. (2020). An empirical analysis of energy intensity and the role of policy instruments. *Energy Policy*, *145*, 111773. https://doi.org/10.1016/j.enpol.2020.111773

Backlund, S., Thollander, P., Palm, J., & Ottosson, M. (2012). Extending the energy efficiency gap. *Energy Policy*, *51*, 392–396. https://doi.org/10.1016/j.enpol.2012.08.042

Badinger, H., & Egger, P. (2015). Productivity Spillovers Across Countries and Industries: New Evidence From OECD Countries. *Oxford Bulletin of Economics and Statistics*, *78*(4), 501–521. https://doi.org/10.1111/obes.12122

Balado Naves, R., Baños-Pino, J., & Mayor, M. (2021). New Insights into the World's Energy Intensity Convergence: A Neoclassical Approach Considering Spatial Spillovers. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3968775

Baldwin, R. (2016). The World Trade Organization and the Future of Multilateralism. *Journal of Economic Perspectives*, *30*(1), 95–116. https://doi.org/10.1257/jep.30.1.95

Baldwin, R., & Lopez-Gonzalez, J. (2015). Supply-chain Trade: A Portrait of Global Patterns and Several Testable Hypotheses. *The World Economy*, *38*(11), 1682–1721. https://doi.org/10.1111/twec.12189

Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute. http://www.jstor.org/stable/j.ctvh4zh1q

Bekun, F. V., Fumey, M. P., Staniewski, M. W., Sun, L., & Agboola, P. O. (2025). Energy intensive growth and the transition pathways: Insights into the role of renewable energy and open market conditions in developing countries. *Energy*, *322*, 135192. https://doi.org/10.1016/j.energy.2025.135192

Berndt, E. R., & Wood, D. O. (1975). Technology, Prices, and the Derived Demand for Energy. *The Review of Economics and Statistics*, *57*(3), 259. https://doi.org/10.2307/1923910

Biswas, R. (2024, January 12). *ASEAN economic outlook in 2024*. S&P Global. https://www.spglobal.com/marketintelligence/en/mi/research-analysis/asean-economic-outlook-in-2024-Jan24.html

Blalock, G., & Gertler, P. J. (2008). Welfare gains from Foreign Direct Investment through technology transfer to local suppliers. *Journal of International Economics*, *74*(2), 402–421. https://doi.org/10.1016/j.jinteco.2007.05.011

Bottazzi, L., & Peri, G. (2003). Innovation and spillovers in regions: Evidence from European patent data. *European Economic Review*, *47*(4), 687–710. https://doi.org/10.1016/s0014-2921(02)00307-0

Britannica. (2024). Industrial Revolution. In *Encyclopedia Britannica*. Britannica. https://www.britannica.com/event/Industrial-Revolution

Brunnermeier, S. B., & Cohen, M. A. (2003). Determinants of environmental innovation in US manufacturing industries. *Journal of Environmental Economics and Management*, *45*(2), 278–293. https://doi.org/10.1016/s0095-0696(02)00058-x

Castaldi, C., Frenken, K., & Los, B. (2014). Related Variety, Unrelated Variety and Technological Breakthroughs: An analysis of US State-Level Patenting. *Regional Studies*, *49*(5), 767–781. https://doi.org/10.1080/00343404.2014.940305

Chang, Y. S., Jeon, S., & Park, N. H. (2019). Do time phase and income influence the convergence in energy intensity A cross-country analysis. *International Journal of Energy Technology and Policy*, *15*(2/3), 301. https://doi.org/10.1504/ijetp.2019.098955

Chintrakarn, P. (2013). Subnational trade flows and state-level energy intensity: an empirical analysis. *Applied Economics Letters*, *20*(14), 1344–1351. https://doi.org/10.1080/13504851.2013.806773

Coe, D. T., & Helpman, E. (1995). International R&D spillovers. *European Economic Review*, *39*(5), 859–887.

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive Capacity: a New Perspective on Learning and Innovation. *Administrative Science Quarterly*, *35*(1), 128–152. https://doi.org/10.2307/2393553

Copeland, B. R., & Taylor, M. Scott. (1999). Trade, spatial separation, and the environment. *Journal of International Economics*, *47*(1), 137–168. https://doi.org/10.1016/s0022-1996(98)00020-8

Costantini, V., Crespi, F., Marin, G., & Paglialunga, E. (2017). Eco-innovation, sustainable supply chains and environmental performance in European industries. *Journal of Cleaner Production*, *155*, 141–154. https://doi.org/10.1016/j.jclepro.2016.09.038

Costantini, V., Crespi, F., & Palma, A. (2017). Characterizing the policy mix and its impact on eco-innovation: A patent analysis of energy-efficient technologies. *Research Policy*, *46*(4), 799–819. https://doi.org/10.1016/j.respol.2017.02.004

Delera, M., & Foster-McGregor, N. (2023). Revisiting international knowledge spillovers: the role of GVCs. *Industrial and Corporate Change*, *32*(5), 1163–1191. https://doi.org/10.1093/icc/dtad046

Audretsch, D. B., & Feldman, M. P. (1996). R&D Spillovers and the Geography of Innovation and Production. *The American Economic Review*, *86*(3), 630–640. http://www.jstor.org/stable/2118216

Jaffe, A. B. (1989). Real Effects of Academic Research. *The American Economic Review*, *79*(5), 957–970. http://www.jstor.org/stable/1831431

Doytch, N., & Narayan, S. (2016). Does FDI influence renewable energy consumption? An analysis of sectoral FDI impact on renewable and non-renewable industrial energy consumption. *Energy Economics*, *54*, 291–301. https://doi.org/10.1016/j.eneco.2015.12.010

Eder, L., & Provornaya, I. (2018). Analysis of energy intensity trend as a tool for long-term forecasting of energy consumption. *Energy Efficiency*, *11*(8), 1971–1997. https://doi.org/10.1007/s12053-018-9656-2

Evenson, R. and Westphal, L. (1995) Technological Change and Technology Strategy. In: Behrman, J. and Srinivasan, T.N., Eds., Handbook of Development Economics, North-Holland, Amsterdam, 2209-2300.

Filippini, M., & Hunt, L. C. (2015). Measurement of energy efficiency based on economic foundations. *Energy Economics*, *52*, S5–S16. https://doi.org/10.1016/j.eneco.2015.08.023

Fiorito, G. (2013). Can we use the energy intensity indicator to study "decoupling" in modern economies?. *Journal of Cleaner Production*, *47*, 465–473. https://doi.org/10.1016/j.jclepro.2012.12.031

Fisher-Vanden, K., Jefferson, G. H., Liu, H., & Tao, Q. (2004). What is driving China's decline in energy intensity? *Resource and Energy Economics*, *26*(1), 77–97. https://doi.org/10.1016/j.reseneeco.2003.07.002

Galindo-Rueda, F., & Verger, F. (2025). *OECD Taxonomy of Economic Activities Based on R&D Intensity*. OECD. https://dx.doi.org/10.1787/5jlv73sqqp8r-en

Gereffi, G., Humphrey, J., & Sturgeon, T. (2005). The Governance of Global Value Chains. *Review of International Political Economy*, *12*(1), 78–104. https://doi.org/10.1080/09692290500049805

Gereffi, G., Lim, H.-C., & Lee, J. (2021). Trade policies, firm strategies, and adaptive reconfigurations of global value chains. *Journal of International Business Policy*, *4*(1). https://doi.org/10.1057/s42214-021-00102-z

Ghisetti, C., & Quatraro, F. (2017). Green Technologies and Environmental Productivity: A Cross-sectoral Analysis of Direct and Indirect Effects in Italian

Regions. *Ecological Economics*, *132*, 1–13. https://doi.org/10.1016/j.ecolecon.2016.10.003

Girma, S. (2005). Absorptive Capacity and Productivity Spillovers from FDI: A Threshold Regression Analysis*. *Oxford Bulletin of Economics and Statistics*, *67*(3), 281–306. https://doi.org/10.1111/j.1468-0084.2005.00120.x

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik Instruments: What, When, Why, and How. *American Economic Review*, *110*(8), 2586–2624. https://doi.org/10.1257/aer.20181047

Gorg, H., & Greenaway, D. (2004). Much Ado about Nothing? Do Domestic Firms Really Benefit from Foreign Direct Investment? *The World Bank Research Observer*, *19*(2), 171–197. https://doi.org/10.1093/wbro/lkh019

Griliches, Z., 1979. Issues in assessing the contribution of research and development to productivity growth. Bell J. Econ. 10 (1), 92–116. https://doi.org/10.2307/3003321.

Grossman, G. M., & Helpman, E. (1991). Trade, knowledge spillovers, and growth. *European Economic Review*, *35*(2-3), 517–526. https://doi.org/10.1016/0014-2921(91)90153-a

Grossman, G. M., & Krueger, A. B. (1995). Economic Growth and the Environment. *The Quarterly Journal of Economics*, *110*(2), 353–377. https://doi.org/10.2307/2118443

Guo, J., Zou, L.-L., & Wei, Y.-M. (2010). Impact of inter-sectoral trade on national and global CO2 emissions: An empirical analysis of China and US. *Energy Policy*, *38*(3), 1389–1397. https://doi.org/10.1016/j.enpol.2009.11.020

Hauknes, J., & Knell, M. (2009). Embodied knowledge and sectoral linkages: An input–output approach to the interaction of high- and low-tech industries. *Research Policy*, *38*(3), 459–469. https://doi.org/10.1016/j.respol.2008.10.012

Hausmann, R., & Rodrik, D. (2003). Economic development as self-discovery. *Journal of Development Economics*, *72*(2), 603–633. https://doi.org/10.1016/s0304-3878(03)00124-x

Hendrickson, C. T., Lave, L. B., & H Scott Matthews. (2006). *Environmental life cycle assessment of goods and services : an input-output approach*. Resources For The Future.

Hicks, J. R. (1932). Marginal Productivity and the Principle of Variation. *Economica*, *35*, 79. https://doi.org/10.2307/2548977

Hill, H., & Menon, J. (2010). ASEAN Economic Integration: Features, Fulfillments, Failures and the Future. *RePEc: Research Papers in Economics*, *69*.

Hirschman, A. O. (1958). *The Strategy of Economic Development*. Yale University Press.

Hobday, M. (1995). East Asian latecomer firms: Learning the technology of electronics. *World Development*, *23*(7), 1171–1193. https://doi.org/10.1016/0305-750x(95)00035-b

Huang, J., Du, D., & Tao, Q. (2017). An analysis of technological factors and energy intensity in China. *Energy Policy*, *109*, 1–9. https://doi.org/10.1016/j.enpol.2017.06.048

Huang, J., & Yu, S. (2016). Effects of investment on energy intensity: evidence from China. *Chinese Journal of Population Resources and Environment*, *14*(3), 197–207. https://doi.org/10.1080/10042857.2016.1215823

Huang, Y., Fang, Y., & Gu, G. (2022). Innovation Effect and Regional Difference of Related Variety: An Empirical Study Based on Prefecture-level City Data of China. *Applied Spatial Analysis and Policy*. https://doi.org/10.1007/s12061-021-09404-4

Hudson, E. A., & Jorgenson, D. W. (1974). U. S. Energy Policy and Economic Growth, 1975-2000. *The Bell Journal of Economics and Management Science*, *5*(2), 461. https://doi.org/10.2307/3003118

Imbruno, M., & Ketterer, T. D. (2018). Energy efficiency gains from importing intermediate inputs: Firm-level evidence from Indonesia. *Journal of Development Economics*, *135*, 117–141. https://doi.org/10.1016/j.jdeveco.2018.06.014

Intarakumnerd, P., Chairatana, P., & Tangchitpiboon, T. (2002). National innovation system in less successful developing countries: the case of Thailand. *Research Policy*, *31*(8-9), 1445–1457. https://doi.org/10.1016/s0048-7333(02)00074-4

Islam, M. T., & Chadee, D. (2021). Stuck at the bottom: Role of tacit and explicit knowledge on innovation of developing-country suppliers in global value chains. *International Business Review*, *32*(2), 101898. https://doi.org/10.1016/j.ibusrev.2021.101898

Jaffe, A. B., Newell, R. G., & Stavins, R. N. (2005). A tale of two market failures: Technology and environmental policy. *Ecological Economics*, *54*(2-3), 164–174. https://doi.org/10.1016/j.ecolecon.2004.12.027

Jaffe, A. B., & Stavins, R. N. (1994). The energy-efficiency gap What does it mean? *Energy Policy*, *22*(10), 804–810. https://doi.org/10.1016/0301-4215(94)90138-4

Jaffe, A. B., Trajtenberg, M., & Henderson, R. (1993). Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations. *The Quarterly Journal of Economics*, *108*(3), 577–598. https://doi.org/10.2307/2118401

Javorcik, B. S. (2004). Does Foreign Direct Investment Increase the Productivity of Domestic Firms? In Search of Spillovers Through Backward Linkages. *American Economic Review*, *94*(3), 605–627. https://doi.org/10.1257/0002828041464605

Jiao, J., Chen, C., & Bai, Y. (2020). Is green technology vertical spillovers more significant in mitigating carbon intensity? Evidence from Chinese industries. *Journal of Cleaner Production*, *257*, 120354. https://doi.org/10.1016/j.jclepro.2020.120354

Johansson, M. T., & Thollander, P. (2018). A review of barriers to and driving forces for improved energy efficiency in Swedish industry– Recommendations for successful in-house energy management. *Renewable and Sustainable Energy Reviews*, *82*, 618–628. https://doi.org/10.1016/j.rser.2017.09.052

Karimu, A., Brännlund, R., Lundgren, T., & Söderholm, P. (2017). Energy intensity and convergence in Swedish industry: A combined econometric and decomposition analysis. *Energy Economics*, *62*, 347–356. https://doi.org/10.1016/j.eneco.2016.07.017

Kaya, Y. (1990). *Impact of Carbon Dioxide Emission on GNP Growth: Interpretation of Proposed Scenarios*. Paris: IPCC/Response Strategies Working Group.

Kazmaganbetova, M., Suleimenov, B., Ayashev, K., & Kerimray, A. (2016). Sectoral structure and energy use in Kazakhstan's regions. *4th IET Clean Energy and Technology Conference (CEAT 2016)*, 9 (7 .)-9 (7 .). https://doi.org/10.1049/cp.2016.1266

Keller, W. (2002). Geographical Localization of International Technology Diffusion. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.217074

Keller, W. (2004). International Technology Diffusion. *Journal of Economic Literature*, *42*(3), 752–782. https://doi.org/10.1257/0022051042177685

Khatib, H. (1995). Energy intensity: a new look. *Energy Policy*, *23*(8), 727–729. https://doi.org/10.1016/0301-4215(95)00026-f

Kuswardana, I., Nachrowi, N. D., Falianty, T. A. F., & Damayanti, A. (2021). The effect of knowledge spillover on productivity: Evidence from manufacturing industry in Indonesia. *ProQuest*. https://doi.org/10.1080/23322039.2021.1923882

Lall, S. (1992). Technological Capabilities and Industrialization. *World Development*, *20*(2), 165–186.

Lall, S. (2000). The Technological Structure and Performance of Developing Country Manufactured Exports, 1985‑98. *Oxford Development Studies*, *28*(3), 337–369. https://doi.org/10.1080/713688318

Landes, D. S. (1969). The Unbound Prometheus: Technological Change and Industrial Development in Western Europe from 1750 to the Present. Cambridge University Press.

Liang, W., Yu, W., & Yao, X. (2025). Liberalization of upstream productive services and green innovation in downstream manufacturing firms: Evidence from China. *Energy Economics*, 108173–108173. https://doi.org/10.1016/j.eneco.2024.108173

Linn, J. (2008). Energy Prices and the Adoption of Energy-Saving Technology*. *The Economic Journal*, *118*(533), 1986–2012. https://doi.org/10.1111/j.1468-0297.2008.02199.x

Liu, W., Xu, X., Yang, Z., Zhao, J., & Xing, J. (2016). Impacts of FDI Renewable Energy Technology Spillover on China's Energy Industry Performance. *Sustainability*, *8*(9), 846. https://doi.org/10.3390/su8090846

Long, R., Wang, J., Chen, H., Li, Q., Wu, M., & Tan-Soo, J.-S. (2023). Applying multilevel structural equation modeling to energy-saving behavior: The interaction of individual- and city-level factors. *Energy Policy*, *174*, 113423. https://doi.org/10.1016/j.enpol.2023.113423

Lundvall, B.-Å. (1988). Innovation as an Interactive Process: From User Producer Interaction to National systems of Innovation. In Dosi, G. et. al. (eds.) (Ed.), *Technical Change and Economic theory*

Ma, C., & Stern, D. I. (2008). China's changing energy intensity trend: A decomposition analysis. *Energy Economics*, *30*(3), 1037–1053. https://doi.org/10.1016/j.eneco.2007.05.005

Ma, Y., Zhao, Y., Jia, R., Wang, W., & Zhang, B. (2022). Impact of financial development on the energy intensity of developing countries. *Heliyon*, *8*(8), e09904. https://doi.org/10.1016/j.heliyon.2022.e09904

Madina Kozhukhova, Botagoz Amanzholova, & Miras Zhiyenbayev. (2019). THE LEGAL REGULATION OF ENERGY EFFICIENCY AND ENERGY SAVING POLICIES IN THE REPUBLIC OF KAZAKHSTAN. *International Journal of Energy Economics and Policy*, *9*(4), 54–62. https://doi.org/10.32479/ijeep.7550

Marchi, N., Bazarian, J. J., Puvenna, V., Janigro, M., Ghosh, C., Zhong, J., Zhu, T., Blackman, E., Stewart, D., Ellis, J., Butler, R., & Janigro, D. (2013). Consequences of Repeated Blood-Brain Barrier Disruption in Football Players. *PLoS ONE*, *8*(3), e56805. https://doi.org/10.1371/journal.pone.0056805

Marshall, A. (1890). *Principles of Economics*.

Martin, R., Muûls, M., de Preux, L. B., & Wagner, U. J. (2012). Anatomy of a paradox: Management practices, organizational structure and energy efficiency. *Journal of Environmental Economics and Management*, *63*(2), 208–223. https://doi.org/10.1016/j.jeem.2011.08.003

Metcalf, G. E. (2008). An Empirical Analysis of Energy Intensity and Its Determinants at the State Level. *The Energy Journal*, *29*(3). https://doi.org/10.5547/issn0195-6574-ej-vol29-no3-1

Miketa, A. (2001). Analysis of energy intensity developments in manufacturing sectors in industrialized and developing countries. *Energy Policy*, *29*(10), 769–775. https://doi.org/10.1016/s0301-4215(01)00010-6

Mokyr, J. (1990). The Lever of Riches: Technological Creativity and Economic Progress. Oxford University Press.

Mulder, P., & de Groot, H. L. F. (2012). Structural change and convergence of energy intensity across OECD countries, 1970–2005. *Energy Economics*, *34*(6), 1910–1921. https://doi.org/10.1016/j.eneco.2012.07.023

Shi, X., 2015. Energy Efficiencies in ASEAN Region, in Yan, J. (eds), The Handbook of Clean Energy Systems (Volume 6), Wiley-Blackwell, 6: 3681-3699

Nelson, R. R., & Winter, S. G. (1982). An Evolutionary Theory of Economic Change. *Southern Economic Journal*, *50*(2), 590. https://doi.org/10.2307/1058233

Newell, R. G., Jaffe, A. B., & Stavins, R. N. (1998). The Induced Innovation Hypothesis and Energy-Saving Technological Change. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.223777

OECD/IEA. (2013). *ENERGY OUTLOOK*. https://www.ctc-n.org/sites/www.ctc-n.org/files/resources/southeastasiaenergyoutlook_weo2013specialreport.pdf

Pane, D. D., & Patunru, A. A. (2022). The role of imported inputs in firms' productivity and exports: evidence from Indonesia. *Review of World Economics*. https://doi.org/10.1007/s10290-022-00476-z

Petrović, P., Filipović, S., & Radovanović, M. (2018). Underlying causal factors of the European Union energy intensity: Econometric evidence. *Renewable and Sustainable Energy Reviews*, *89*, 216–227. https://doi.org/10.1016/j.rser.2018.03.061

Pindyck, R. S. (1979). Interfuel Substitution and the Industrial Demand for Energy: An International Comparison. *The Review of Economics and Statistics*, *61*(2), 169. https://doi.org/10.2307/1924584

Popp, D. (2002). Induced Innovation and Energy Prices. *American Economic Review*, *92*(1), 160–180. https://doi.org/10.1257/000282802760015658

Popp, D. C. (2001). The effect of new technology on energy consumption. *Resource and Energy Economics*, *23*(3), 215–239. https://doi.org/10.1016/s0928-7655(00)00045-2

Popp, D., Newell, R. G., & Jaffe, A. B. (2010). Energy, the Environment, and Technological Change. *Handbook of the Economics of Innovation, Volume 2*, *2*, 873–937. https://doi.org/10.1016/s0169-7218(10)02005-8

Proskuryakova, L., & Kovalev, A. (2015). Measuring energy efficiency: Is energy intensity a good evidence base? *Applied Energy*, *138*, 450–459. https://doi.org/10.1016/j.apenergy.2014.10.060

Radovanović, M., & Filipović, S. (2015). New Approach to Energy Intensity in the Eu – Total Energy and Carbon Cost Approach. *Energy & Environment*, *26*(4), 601–616. https://doi.org/10.1260/0958-305x.26.4.601

Rahko, J. (2025). Vertical spillovers and the energy intensity of European industries. *Energy Economics*, *141*, 108053. https://doi.org/10.1016/j.eneco.2024.108053

Rhisa Azaliah, Kurniawan, H., Hartono, D., & Putu Angga Widyastaman. (2023). The convergence of energy intensity in developing countries: a spatial econometric analysis with Indonesia's provincial panel data. *Environment Development and Sustainability*, *26*(6), 14915–14939. https://doi.org/10.1007/s10668-023-03227-8

Rockström, J. (2009). A Safe Operating Space for Humanity. *Nature*, *461*(7263), 472–475. https://doi.org/10.1038/461472a

Rodrik, D. (2014). Green industrial policy. *Oxford Review of Economic Policy*, *30*(3), 469–491. https://doi.org/10.1093/oxrep/gru025

Romer, P. M. (1986). Increasing returns and long-run growth. *Journal of Political Economy*, *94*(5), 1002–1037. https://doi.org/10.1086/261420

Samuelson, R. D. (2014). The unexpected challenges of using energy intensity as a policy objective: Examining the debate over the APEC energy intensity goal. *Energy Policy*, *64*, 373–381. https://doi.org/10.1016/j.enpol.2013.09.020

Saraswati, A., Hartono, D., & Indriyani, W. (2022). The impact of FDI on energy intensity: a spatial econometric analysis of Indonesian provinces. *Journal of Environmental Studies and Sciences*. https://doi.org/10.1007/s13412-022-00775-9

Scherer, F. M. (1982). Inter-industry technology flows in the United States. *Research Policy*, *11*(4), 227–245. https://doi.org/10.1016/0048-7333(82)90011-7

Scherer, K. R. (1982). Emotion as a process: Function, origin and regulation. *Social Science Information*, *21*(4-5), 555–570. https://doi.org/10.1177/053901882021004004

Schmookler, J. (1966). *Invention and Economic Growth*. https://doi.org/10.4159/harvard.9780674432833

Serrano-Domingo, G., & Cabrer-Borrás, B. (2016). Direct and indirect knowledge spillovers and industrial productivity. *Industry and Innovation*, *24*(2), 165–189. https://doi.org/10.1080/13662716.2016.1224706

Setyawan, D. (2020). Global Journal of Environmental Science and Management Economy-wide energy efficiency using a comprehensive decomposition method NUMBER OF REFERENCES 41 NUMBER OF FIGURES 3 NUMBER OF TABLES 3. *Global J. Environ. Sci. Manage*, *6*(3), 385–402. https://doi.org/10.22034/gjesm.2020.03.08

Shahbaz, M., Sinha, A., Raghutla, C., & Vo, X. V. (2022). Decomposing scale and technique effects of financial development and foreign direct investment on renewable energy consumption. *Energy*, *238*, 121758. https://doi.org/10.1016/j.energy.2021.121758

Shakya, S. R., Adhikari, R., Poudel, S., & Rupakheti, M. (2022). Energy equity as a major driver of energy intensity in South Asia. *Renewable and Sustainable Energy Reviews*, *170*, 112994. https://doi.org/10.1016/j.rser.2022.112994

Solarin, S. A. (2020). Towards sustainable development in developing countries: Aggregate and disaggregate analysis of energy intensity and the role of fossil

fuel subsidies. *Sustainable Production and Consumption*, *24*, 254–265. https://doi.org/10.1016/j.spc.2020.07.011

Southeast Asia Energy Outlook 2017. (2017). IEA.

Staiger, D., & Stock, J. H. (1997). Instrumental Variables Regression with Weak Instruments. *Econometrica*, *65*(3), 557. https://doi.org/10.2307/2171753

Stavins, R. N., Jaffe, A. B., & Newell, R. G. (2002). Environmental Policy and Technological Change. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.311023

Stern, D. (2025). *Economic growth and energy*. The Australian National University; Elsevier. https://researchportalplus.anu.edu.au/en/publications/economic-growth-and-energy/

Stock, J. H., & Yogo, M. (2005). Testing for Weak Instruments in Linear IV Regression. *Identification and Inference for Econometric Models*, 80–108. https://doi.org/10.1017/cbo9780511614491.006

Sue Wing, I. (2008). Explaining the declining energy intensity of the U.S. economy. *Resource and Energy Economics*, *30*(1), 21–49. https://doi.org/10.1016/j.reseneeco.2007.03.001

Sun, H., Edziah, B. K., Sun, C., & Kporsu, A. K. (2019). Institutional quality, green innovation and energy efficiency. *Energy Policy*, *135*, 111002. https://doi.org/10.1016/j.enpol.2019.111002

Taylor, M. S., Grossman, G. M., & Helpman, E. (1993). Innovation and Growth in the Global Economy. *Economica*, *60*(239), 373. https://doi.org/10.2307/2554862

Terleckyj, N. (1974). *Effects of R&D on the Productivity Growth of Industries: An Explanatory Study*. National Planning Association.

Terleckyj, N. E. (1974). Effects of R&D on the productivity growth of industries: an exploratory study. National Planning Association.

Thaler, D. R. (2011). The determinants of energy intensity. University of Florida

Todo, Y., Matous, P., & Inoue, H. (2016). The strength of long ties and the weakness of strong ties: Knowledge diffusion through supply chain networks. *Research Policy*, *45*(9), 1890–1906. https://doi.org/10.1016/j.respol.2016.06.008

Tsemekidi Tzeiranaki, S., Bertoldi, P., Castellazzi, L., Gonzalez Torres, M., Clementi, E., Paci, D., 2022. Energy Consumption and Energy Efficiency Trends in the EU, 2000- 2020. Publications Office of the European Union. https://doi.org/10.2760/727548.

United Nations. (2001). *INTENSITY OF ENERGY USE*. Un.org. https://www.un.org/esa/sustdev/natlinfo/indicators/isdms2001/isd-ms2001economicB.htm?utm_source

Verbič, M., Filipović, S., & Radovanović, M. (2017). Electricity prices and energy intensity in Europe. *Utilities Policy*, *47*, 58–68. https://doi.org/10.1016/j.jup.2017.07.001

Verdolini, E., & Galeotti, M. (2011). At home and abroad: An empirical analysis of innovation and diffusion in energy technologies. *Journal of Environmental Economics and Management*, *61*(2), 119–134. https://doi.org/10.1016/j.jeem.2010.08.004

Voigt, S., De Cian, E., Schymura, M., & Verdolini, E. (2014). Energy intensity developments in 40 major economies: Structural change or technology

improvement? *Energy Economics*, *41*, 47–62. https://doi.org/10.1016/j.eneco.2013.10.015

s *Energy Economics*, *48*, 253–264. https://doi.org/10.1016/j.eneco.2014.12.014

Wang, J., Chen, K., & Song, X. (2020). Differences Among Influencing Factors of China's Provincial Energy Intensity: Empirical Analysis from a Geographically Weighted Regression Model. *Polish Journal of Environmental Studies*, *29*(4), 2901–2916. https://doi.org/10.15244/pjoes/113097

Ward, J. D., Sutton, P. C., Werner, A. D., Costanza, R., Mohr, S. H., & Simmons, C. T. (2016). Is Decoupling GDP Growth from Environmental Impact Possible? *PLOS ONE*, *11*(10), e0164733. https://doi.org/10.1371/journal.pone.0164733

Wolff, E. (2011). *Spillovers, Linkages, and Productivity Growth in the US Economy, 1958 to 2007*. https://doi.org/10.3386/w16864

Wrigley, E. A. (2010). *Energy and the English Industrial Revolution*. https://doi.org/10.1017/cbo9780511779619

Wurlod, J.-D., & Noailly, J. (2018). The impact of green innovation on energy intensity: An empirical analysis for 14 industrial sectors in OECD countries. *Energy Economics*, *71*, 47–61. https://doi.org/10.1016/j.eneco.2017.12.012

Xin, L., Sun, H., & Xia, X. (2022). Renewable energy technology innovation and inclusive low-carbon development from the perspective of spatiotemporal consistency. *Environmental Science and Pollution Research*, *30*(8), 20490–20513. https://doi.org/10.1007/s11356-022-23556-x

Yang, X., Liu, J., & Li, M. (2024). Transportation energy consumption decreases despite urban sprawl: Evidence from Chinese household survey. *Cities*, *146*, 104738–104738. https://doi.org/10.1016/j.cities.2023.104738

Zhang, F. (2013). The energy transition of the transition economies: An empirical analysis. *Energy Economics*, *40*, 679–686. https://doi.org/10.1016/j.eneco.2013.09.001